

Asymptotic Fisher information for Poisson counts driven by a stationary Gaussian process

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This memo rigorously proves Eq. (3.17) in Ng *et al.*, Phys. Rev. A **93**, 042121 (2016) for the asymptotic Fisher information of spectral photon counting. Let X be a real zero-mean stationary Gaussian process with spectral density S_θ , and let the observations be conditionally independent Poisson random variables whose means are the squared moduli of the positive-frequency discrete Fourier coefficients of a length- T sample of X . Under Wiener-algebra regularity and strict positivity of the spectrum, we prove

$$\lim_{T \rightarrow \infty} \frac{I_T(\theta)}{T} = \frac{1}{2\pi} \int_0^\pi \frac{[\partial_\theta S_\theta(\omega)]^2}{S_\theta(\omega)[1 + S_\theta(\omega)]} d\omega.$$

ChatGPT 5.6 Sol produced the proof, following a suggestion by Tsang. Claude Opus 5.5 checked the proof and proposed minor improvements.

I. INTRODUCTION

The goal of this work is to rigorously prove Eq. (3.17) in Ref. [1] for the asymptotic Fisher information of spectral photon counting, after a zero-mean stationary Gaussian process with a parameter-dependent spectral density modulates a coherent optical beam.

Ref. [1] derives the formula by showing that the photon counts tend to a set of independent Bose–Einstein (geometric) random variables in the long-time limit and then computing the Fisher information of the Bose–Einstein variables. However, ChatGPT complained that this approach is not rigorous, as one should instead take the long-time limit of the Fisher information to determine the asymptotic Cramér–Rao bound; taking the long-time limit prematurely need not give the correct result.

Following Refs. [1, 2], Tsang suggested to ChatGPT the general strategy of sandwiching the Fisher information between a lower bound due to Stein, Mezghani, and Nossek [3] and an upper bound due to Zamir [4], as he suspected that the two bounds, once optimized, should give the same asymptotic formula. ChatGPT took the suggestion and eventually produced the proof below, after a few rounds of encouragement by Tsang. The proof was then bounced between Claude and ChatGPT for checking and improvements. Both eventually claimed the proof to be correct, although Tsang has not checked it independently.

II. MODEL AND ASSUMPTIONS

Identify $\mathbb{T}$ with $[-\pi, \pi)$ and let $A(\mathbb{T})$ denote the Wiener algebra

$$A(\mathbb{T}) = \left\{ a(\omega) = \sum_{k \in \mathbb{Z}} a_k e^{-ik\omega} : \sum_{k \in \mathbb{Z}} |a_k| < \infty \right\}, \quad \|a\|_A = \sum_k |a_k|.$$

Fix a parameter value θ_0 . Assume that on an open interval Θ containing θ_0 :

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(A1) $\theta \mapsto S_\theta$ is C^1 as a map $\Theta \rightarrow A(\mathbb{T})$;

(A2) S_θ is real and even, and there are constants $0 < m < M < \infty$ such that

$$m \leq S_\theta(\omega) \leq M, \quad \theta \in \Theta, \omega \in \mathbb{T}.$$

No separate likelihood-regularity assumption is imposed: the finite-dimensional Gaussian and Gaussian–Poisson models used below are shown in Appendix A to possess the score identities and finite Fisher information required by the proof.

Let $X = (X_t)_{t \in \mathbb{Z}}$ be a real, zero-mean stationary Gaussian process with spectral density S_θ . Thus

$$C_\theta(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_\theta(\omega) e^{ik\omega} d\omega, \quad S_\theta(\omega) = \sum_{k \in \mathbb{Z}} C_\theta(k) e^{-ik\omega}.$$

For $T \geq 3$ let $\omega_j = 2\pi j/T$ and let F_T be the unitary DFT matrix

$$(F_T)_{jt} = T^{-1/2} e^{-i\omega_j t}, \quad j = 0, \dots, T-1, \quad t = 1, \dots, T.$$

For $X^{(T)} = (X_1, \dots, X_T)^\top$ put $\check{X}_T = F_T X^{(T)}$, that is,

$$\check{X}_{T,j} = \frac{1}{\sqrt{T}} \sum_{t=1}^T X_t e^{-i\omega_j t}, \quad j = 0, \dots, T-1,$$

and let $J_T = \{j : 0 < \omega_j < \pi\}$. Conditional on $X^{(T)}$, the observations are independent and

$$N_{T,j} \mid X^{(T)} \sim \text{Pois}(|\check{X}_{T,j}|^2), \quad j \in J_T. \quad (2.1)$$

Write $I_T(\theta)$ for the Fisher information in $N_T = (N_{T,j})_{j \in J_T}$.

Theorem 1. *Under (A1)–(A2),*

$$\boxed{\lim_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} = \frac{1}{2\pi} \int_0^\pi \frac{\dot{S}(\omega)^2}{S(\omega)[1 + S(\omega)]} d\omega,} \quad (2.2)$$

where $S = S_{\theta_0}$ and $\dot{S} = \partial_\theta S_\theta|_{\theta_0}$.

Theorem 1 leads to Eq. (3.17) in Ref. [1] after some trivial rescaling.

We write $\mathcal{I}$ for the right-hand side of (2.2). Throughout, a dot denotes ∂_θ , and quantities written without a θ subscript are evaluated at θ_0 . Theorem 1 follows from Proposition 1 (Section IV) and Proposition 2 (Section V).

III. TOEPLITZ–CIRCULANT LEMMAS

For $a \in A(\mathbb{T})$ define the Toeplitz matrix $\text{Toe}_T(a) = (a_{r-s})_{r,s=1}^T$, so that $\text{Cov}(X^{(T)}) = \text{Toe}_T(S_\theta)$. The map $a \mapsto \text{Toe}_T(a)$ is linear, $\text{Toe}_T(1) = I$, and $\text{Toe}_T(a)$ is Hermitian when a is real. For $v \in \mathbb{C}^T$,

$$v^\dagger \text{Toe}_T(a) v = \frac{1}{2\pi} \int_{-\pi}^{\pi} a(\omega) \left| \sum_{r=1}^T v_r e^{-ir\omega} \right|^2 d\omega, \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| \sum_{r=1}^T v_r e^{-ir\omega} \right|^2 d\omega = \|v\|^2, \quad (3.1)$$

so $(\min a)I \leq \text{Toe}_T(a) \leq (\max a)I$ for real a . In particular, $mI \leq \text{Toe}_T(S_\theta) \leq MI$ by (A2).

For a bounded function a on $\mathbb{T}$ define the sampled circulant

$$\text{Cir}_T(a) = F_T^\dagger \text{diag}(a(\omega_0), \dots, a(\omega_{T-1})) F_T.$$

Sampled circulants are simultaneously diagonalized by F_T , so $\text{Cir}_T(a) \text{Cir}_T(b) = \text{Cir}_T(ab)$, $\text{Cir}_T(a)^\dagger = \text{Cir}_T(a^*)$, $\|\text{Cir}_T(a)\|_{\text{op}} = \max_j |a(\omega_j)|$, and $f(\text{Cir}_T(a)) = \text{Cir}_T(f \circ a)$ for every function f defined on the values $a(\omega_j)$; in particular $\text{Cir}_T(a)^{-1} = \text{Cir}_T(1/a)$ if a does not vanish [5, Theorem 7 and Lemma 10].

For $T \times T$ matrices set $\|M\|_T = T^{-1/2} \|M\|_{\text{HS}}$, where $\|M\|_{\text{HS}} = [\text{tr}(M^\dagger M)]^{1/2}$ denotes the Hilbert–Schmidt norm, also called the Frobenius norm. Both $\|\cdot\|_{\text{HS}}$ and $\|\cdot\|_T$ are unitarily invariant, and for all matrices M, M' and every orthogonal projection P ,

$$\begin{aligned} \|M\|_T &\leq \|M\|_{\text{op}}, & \frac{|\text{tr}(PM)|}{T} &\leq \|M\|_T, \\ \|MM'\|_T &\leq \min\{\|M\|_T \|M'\|_{\text{op}}, \|M\|_{\text{op}} \|M'\|_T\}. \end{aligned} \quad (3.2)$$

(The trace bound follows from $|\text{tr}(PM)| \leq \|P\|_{\text{HS}} \|M\|_{\text{HS}}$ and $\|P\|_{\text{HS}} \leq T^{1/2}$.) We write $M_T \sim \widetilde{M}_T$ if $\|M_T - \widetilde{M}_T\|_T \rightarrow 0$ and both sequences have uniformly bounded operator norm [5]. Finally, for continuous g on $\mathbb{T}$ the Riemann-sum theorem gives

$$\frac{1}{T} \sum_{j=0}^{T-1} g(\omega_j) \rightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\omega) d\omega, \quad \frac{1}{T} \sum_{j \in J_T} g(\omega_j) \rightarrow \frac{1}{2\pi} \int_0^\pi g(\omega) d\omega. \quad (3.3)$$

Lemma 1 (Toeplitz–circulant equivalence [5, Lemma 11]). *If $a \in A(\mathbb{T})$, then $\text{Toe}_T(a) \sim \text{Cir}_T(a)$. Moreover, both $\|\text{Toe}_T(a)\|_{\text{op}}$ and $\|\text{Cir}_T(a)\|_{\text{op}}$ are at most $\|a\|_A$.*

Lemma 2 (Stability [5, Theorem 1]). *Suppose $M_T \sim \widetilde{M}_T$ and $M'_T \sim \widetilde{M}'_T$. Then $M_T + M'_T \sim \widetilde{M}_T + \widetilde{M}'_T$, $M_T^\dagger \sim \widetilde{M}_T^\dagger$, and $M_T M'_T \sim \widetilde{M}_T \widetilde{M}'_T$. If, in addition, M_T and $\widetilde{M}_T$ are invertible and their inverses have uniformly bounded operator norm, then $M_T^{-1} \sim \widetilde{M}_T^{-1}$.*

Lemma 3 (Functional calculus for Toeplitz–circulant sequences). *Let $\theta \mapsto a_\theta \in A(\mathbb{T})$ be C^1 in a neighborhood of θ_0 , with a_θ real-valued, and suppose that for some fixed $\alpha < \beta$,*

$$\alpha \leq a_\theta(\omega) \leq \beta$$

for all ω and all θ in that neighborhood. Write

$$a = a_{\theta_0}, \quad \dot{a} = \partial_\theta a_\theta|_{\theta=\theta_0}.$$

Let f be analytic on an open neighborhood of $[\alpha, \beta]$. Then

$$f(\text{Toe}_T(a)) \sim \text{Cir}_T(f(a)), \quad (3.4)$$

and

$$\left. \frac{d}{d\theta} f(\text{Toe}_T(a_\theta)) \right|_{\theta=\theta_0} \sim \text{Cir}_T(f'(a)\dot{a}). \quad (3.5)$$

Proof. By Eq. (3.1) and the eigenvalue representation of circulants, $\text{Toe}_T(a_\theta)$ and $\text{Cir}_T(a_\theta)$ are Hermitian with spectra in $[\alpha, \beta]$ for all θ in the neighborhood, and $\text{Toe}_T(a) \sim \text{Cir}_T(a)$ and $\text{Toe}_T(\dot{a}) \sim \text{Cir}_T(\dot{a})$ by Lemma 1. Since f is continuous on $[\alpha, \beta]$, Theorem 1 of Ref. [6] gives $f(\text{Toe}_T(a)) \sim f(\text{Cir}_T(a)) = \text{Cir}_T(f(a))$, which is Eq. (3.4).

For Eq. (3.5), let γ be a positively oriented simple closed contour enclosing $[\alpha, \beta]$ such that f is analytic on and inside γ , and let $\delta = \text{dist}(\gamma, [\alpha, \beta]) > 0$. If A is Hermitian with spectrum in $[\alpha, \beta]$, Cauchy's integral formula applied to its eigenvalues gives

$$f(A) = \frac{1}{2\pi i} \oint_{\gamma} f(z)(zI - A)^{-1} dz, \quad \|(zI - A)^{-1}\|_{\text{op}} \leq \delta^{-1} \quad (z \in \gamma).$$

Since Toe_T is linear with $\|\text{Toe}_T(b)\|_{\text{op}} \leq \|b\|_A$, the map $\theta \mapsto \text{Toe}_T(a_\theta)$ is C^1 in operator norm, with derivative $\text{Toe}_T(\dot{a})$ at θ_0 . Taking $A = \text{Toe}_T(a_\theta)$ and differentiating under the integral sign therefore gives

$$\left. \frac{d}{d\theta} f(\text{Toe}_T(a_\theta)) \right|_{\theta=\theta_0} = \frac{1}{2\pi i} \oint_{\gamma} f(z) \mathcal{R}_T(z) \text{Toe}_T(\dot{a}) \mathcal{R}_T(z) dz, \quad \mathcal{R}_T(z) = (zI - \text{Toe}_T(a))^{-1}.$$

Likewise, with $\tilde{\mathcal{R}}_T(z) = (zI - \text{Cir}_T(a))^{-1}$, simultaneous DFT diagonalization and Cauchy's formula for f' give

$$\text{Cir}_T(f'(a)\dot{a}) = \frac{1}{2\pi i} \oint_{\gamma} f(z) \tilde{\mathcal{R}}_T(z) \text{Cir}_T(\dot{a}) \tilde{\mathcal{R}}_T(z) dz.$$

The resolvent identity

$$\mathcal{R}_T(z) - \tilde{\mathcal{R}}_T(z) = \mathcal{R}_T(z)(\text{Toe}_T(a) - \text{Cir}_T(a))\tilde{\mathcal{R}}_T(z)$$

and Eq. (3.2) imply

$$\sup_{z \in \gamma} \|\mathcal{R}_T(z) - \tilde{\mathcal{R}}_T(z)\|_T \leq \delta^{-2} \|\text{Toe}_T(a) - \text{Cir}_T(a)\|_T \longrightarrow 0. \quad (3.6)$$

Put

$$E_T = \text{Toe}_T(\dot{a}), \quad \tilde{E}_T = \text{Cir}_T(\dot{a}).$$

By Lemma 1, $\|E_T - \tilde{E}_T\|_T \rightarrow 0$, while

$$\|E_T\|_{\text{op}}, \|\tilde{E}_T\|_{\text{op}} \leq \|\dot{a}\|_A.$$

For every $z \in \gamma$, expand the difference of the two integrands as

$$\begin{aligned} & \mathcal{R}_T E_T \mathcal{R}_T - \tilde{\mathcal{R}}_T \tilde{E}_T \tilde{\mathcal{R}}_T \\ &= (\mathcal{R}_T - \tilde{\mathcal{R}}_T) E_T \mathcal{R}_T + \tilde{\mathcal{R}}_T (E_T - \tilde{E}_T) \mathcal{R}_T + \tilde{\mathcal{R}}_T \tilde{E}_T (\mathcal{R}_T - \tilde{\mathcal{R}}_T), \end{aligned}$$

where the argument z has been suppressed. Using Eq. (3.2) and the uniform resolvent bound gives

$$\begin{aligned} & \|\mathcal{R}_T E_T \mathcal{R}_T - \tilde{\mathcal{R}}_T \tilde{E}_T \tilde{\mathcal{R}}_T\|_T \\ & \leq 2\delta^{-1} \|\dot{a}\|_A \|\mathcal{R}_T - \tilde{\mathcal{R}}_T\|_T + \delta^{-2} \|E_T - \tilde{E}_T\|_T. \end{aligned}$$

The right-hand side tends to zero uniformly for $z \in \gamma$ by (3.6). Consequently,

$$\begin{aligned} & \left\| \left. \frac{d}{d\theta} f(\text{Toe}_T(a_\theta)) \right|_{\theta=\theta_0} - \text{Cir}_T(f'(a)\dot{a}) \right\|_T \\ & \leq \frac{\text{length}(\gamma)}{2\pi} \sup_{z \in \gamma} |f(z)| \sup_{z \in \gamma} \|\mathcal{R}_T E_T \mathcal{R}_T - \tilde{\mathcal{R}}_T \tilde{E}_T \tilde{\mathcal{R}}_T\|_T \longrightarrow 0. \end{aligned}$$

Finally, each integrand has operator norm at most $\delta^{-2} \|\dot{a}\|_A$. The two matrix sequences in Eq. (3.5) therefore have uniformly bounded operator norm, so the preceding convergence is precisely the asserted asymptotic equivalence. $\square$

IV. LOWER BOUND

Set

$$\Gamma_T = \text{Toe}_T(S), \quad A_T = F_T \Gamma_T F_T^\dagger, \quad B_T = F_T \Gamma_T F_T^\top.$$

Since $\check{X}_T = F_T X^{(T)}$,

$$(A_T)_{jk} = \mathbb{E}(\check{X}_{T,j} \check{X}_{T,k}^*), \quad (B_T)_{jk} = \mathbb{E}(\check{X}_{T,j} \check{X}_{T,k}).$$

Write $s_{T,j} = (A_T)_{jj} = \mathbb{E}|\check{X}_{T,j}|^2$, and note that $0 < s_{T,j} \leq M$ because $mI \leq \Gamma_T \leq MI$.

The real and imaginary parts of the $\check{X}_{T,j}$ are jointly Gaussian with mean zero, so Isserlis' theorem gives

$$\text{Cov}(|\check{X}_{T,j}|^2, |\check{X}_{T,k}|^2) = |(A_T)_{jk}|^2 + |(B_T)_{jk}|^2.$$

Using conditional independence and the variance formula for a Poisson variable,

$$\text{Cov}(N_{T,j}, N_{T,k}) = \delta_{jk} s_{T,j} + |(A_T)_{jk}|^2 + |(B_T)_{jk}|^2, \quad j, k \in J_T. \quad (4.1)$$

Define

$$w(\omega) = \frac{\dot{S}(\omega)}{S(\omega)[1 + S(\omega)]}, \quad w_j = w(\omega_j). \quad (4.2)$$

By (A1)–(A2), w is bounded and continuous. At the fixed parameter value θ_0 , define the scalar statistic

$$Q_T = \sum_{j \in J_T} w_j N_{T,j}. \quad (4.3)$$

The weights are held fixed when differentiating in θ .

Proposition 1 (Lower bound). *Under (A1)–(A2), $\liminf_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \geq \mathcal{I}$.*

Proof. Consider the information inequality [7]

$$I_T(\theta_0) \geq \frac{[\partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0}]^2}{\text{Var}_{\theta_0}(Q_T)} \quad (4.4)$$

whenever $\text{Var}_{\theta_0}(Q_T) > 0$.

Numerator. We have $\mathbb{E}_\theta Q_T = \sum_{j \in J_T} w_j s_{T,j}(\theta)$ with $s_{T,j}(\theta) = (F_T \text{Toe}_T(S_\theta) F_T^\dagger)_{jj}$, so

$$\partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0} = \sum_{j \in J_T} w_j \dot{s}_{T,j}, \quad \dot{s}_{T,j} = (F_T \text{Toe}_T(\dot{S}) F_T^\dagger)_{jj}.$$

By unitary invariance and Lemma 1 applied to $\dot{S}$,

$$\frac{1}{T} \sum_{j=0}^{T-1} |\dot{s}_{T,j} - \dot{S}(\omega_j)|^2 \leq \|\text{Toe}_T(\dot{S}) - \text{Cir}_T(\dot{S})\|_T^2 \rightarrow 0.$$

Since w is bounded, the Cauchy–Schwarz inequality and (3.3) imply

$$\frac{1}{T} \partial_\theta \mathbb{E}_\theta Q_T |_{\theta_0} \longrightarrow \frac{1}{2\pi} \int_0^\pi w(\omega) \dot{S}(\omega) d\omega = \mathcal{I}. \quad (4.5)$$

Variance. By Lemma 1 and unitary invariance,

$$\sum_{j,k=0}^{T-1} |(A_T)_{jk} - \delta_{jk} S(\omega_j)|^2 = \|\Gamma_T - \text{Cir}_T(S)\|_{\text{HS}}^2 = o(T). \quad (4.6)$$

Hence

$$\sum_{j \neq k} |(A_T)_{jk}|^2 = o(T), \quad \sum_j |s_{T,j} - S(\omega_j)|^2 = o(T). \quad (4.7)$$

Let Π_T be the permutation matrix of $j \mapsto -j \pmod{T}$. A direct computation gives $F_T F_T^\top = \Pi_T$, hence

$$F_T \text{Cir}_T(S) F_T^\top = \text{diag}(S(\omega_j)) \Pi_T.$$

For $j, k \in J_T$ we have $2 \leq j + k \leq T - 1$, so $k \not\equiv -j \pmod{T}$, and the $J_T \times J_T$ block of this matrix vanishes. As $F_T^\top$ is unitary, Lemma 1 gives

$$\sum_{j,k \in J_T} |(B_T)_{jk}|^2 \leq \|\Gamma_T - \text{Cir}_T(S)\|_{\text{HS}}^2 = o(T). \quad (4.8)$$

By (4.1),

$$\text{Var}(Q_T) = \sum_{j \in J_T} w_j^2 (s_{T,j}^2 + s_{T,j}^2) + \sum_{\substack{j,k \in J_T \\ j \neq k}} w_j w_k |(A_T)_{jk}|^2 + \sum_{j,k \in J_T} w_j w_k |(B_T)_{jk}|^2,$$

and by boundedness of w , (4.7) and (4.8), the last two sums are $o(T)$. Since $|s_{T,j}^2 - S(\omega_j)^2| \leq 2M|s_{T,j} - S(\omega_j)|$, the second relation in (4.7), the Cauchy–Schwarz inequality and (3.3) yield

$$\frac{1}{T} \text{Var}(Q_T) \longrightarrow \frac{1}{2\pi} \int_0^\pi w(\omega)^2 S(\omega) [1 + S(\omega)] d\omega = \mathcal{I}. \quad (4.9)$$

If $\mathcal{I} > 0$, then $\text{Var}(Q_T) > 0$ for large T , and (4.4), (4.5) and (4.9) imply

$$\liminf_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \geq \mathcal{I}. \quad (4.10)$$

If $\mathcal{I} = 0$, (4.10) follows from nonnegativity of Fisher information. $\square$

V. UPPER BOUND BY CHANGE OF LATENT VARIABLE

Define, for every $\theta \in \Theta$,

$$R_\theta = 1 + S_\theta, \quad H_\theta = \frac{S_\theta}{1 + S_\theta}. \quad (5.1)$$

Then $S_\theta = R_\theta H_\theta$ and, since $\text{Toe}_T(1) = I$, $\text{Toe}_T(R_\theta) = I + \text{Toe}_T(S_\theta)$. Let

$$Z^{(T)} \sim N(0, \text{Toe}_T(R_\theta))$$

and set

$$L_T(\theta) = \text{Toe}_T(S_\theta)^{1/2} \text{Toe}_T(R_\theta)^{-1/2}, \quad X^{(T)} = L_T(\theta) Z^{(T)}. \quad (5.2)$$

Both factors of $L_T(\theta)$ are functions of the same real symmetric matrix $\text{Toe}_T(S_\theta)$, so

$$L_T(\theta) = h(\text{Toe}_T(S_\theta)), \quad h(x) = \sqrt{\frac{x}{1+x}}. \quad (5.3)$$

In particular $L_T(\theta)$ is real symmetric, and $L_T(\theta) \text{Toe}_T(R_\theta) L_T(\theta)^\top = \text{Toe}_T(S_\theta)$ because $h(x)^2(1+x) = x$. Thus $X^{(T)}$ has exactly the original Gaussian law, and if N_T is generated from $X^{(T)}$ by (2.1), its marginal law is exactly the original law, for every T and every $\theta \in \Theta$. By Lemma A.1 in Appendix A, applied to (5.1),

$$I_T(\theta_0) \leq I_T^Z + \mathbb{E} I_T^{N|Z}(Z^{(T)}), \quad (5.4)$$

where I_T^Z is the Fisher information of $Z^{(T)}$ and $I_T^{N|Z}(z)$ is the conditional Fisher information of N_T given $Z^{(T)} = z$, both at θ_0 .

Proposition 2 (Upper bound). *Under (A1)–(A2), $\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \mathcal{I}$.*

Proof. Gaussian term. For a zero-mean real Gaussian vector whose positive definite covariance G_θ is C^1 in θ , the Fisher information is $\frac{1}{2} \text{tr}(G^{-1} \dot{G} G^{-1} \dot{G})$. Therefore

$$I_T^Z = \frac{1}{2} \text{tr}\{\text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R}) \text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R})\}.$$

By Lemma 1, $\text{Toe}_T(R) \sim \text{Cir}_T(R)$ and $\text{Toe}_T(\dot{R}) \sim \text{Cir}_T(\dot{R})$. Since $\text{Toe}_T(R) \geq (1+m)I$ and $\text{Cir}_T(R) \geq (1+m)I$, Lemma 2 gives $\text{Toe}_T(R)^{-1} \sim \text{Cir}_T(1/R)$, and then that the product inside the trace is asymptotically equivalent to $\text{Cir}_T((\dot{R}/R)^2)$. By (3.2) with $P = I$, normalized traces of asymptotically equivalent matrices differ by $o(1)$, so (3.3) and evenness give

$$\lim_{T \rightarrow \infty} \frac{I_T^Z}{T} = \frac{1}{4\pi} \int_{-\pi}^{\pi} \left(\frac{\dot{R}}{R} \right)^2 d\omega = \frac{1}{2\pi} \int_0^\pi (\partial_\theta \log R)^2 d\omega. \quad (5.5)$$

Conditional Poisson term. At θ_0 put

$$\check{X} = F_T L_T Z^{(T)}, \quad V = F_T \dot{L}_T Z^{(T)},$$

where $\dot{L}_T = \partial_\theta L_T(\theta)|_{\theta_0}$. Conditional on $Z^{(T)}$, the Poisson mean at frequency $j \in J_T$ is $\lambda_j = |\check{X}_j|^2$, which is almost surely positive (see Lemma A.1), and, holding $Z^{(T)}$ fixed,

$$\dot{\lambda}_j = 2 \text{Re}(\check{X}_j^* V_j).$$

The conditional Fisher information is therefore

$$I_T^{N|Z}(Z^{(T)}) = \sum_{j \in J_T} \frac{\dot{\lambda}_j^2}{\lambda_j} = 4 \sum_{j \in J_T} \frac{[\text{Re}(\check{X}_j^* V_j)]^2}{|\check{X}_j|^2},$$

and since $[\text{Re}(y^* v)]^2 \leq |y|^2 |v|^2$,

$$I_T^{N|Z}(Z^{(T)}) \leq 4 \sum_{j \in J_T} |V_j|^2. \quad (5.6)$$

We next identify $\dot{L}_T$. Apply Lemma 3 with

$$a_\theta = S_\theta, \quad f(x) = h(x) = \sqrt{\frac{x}{1+x}}.$$

By (A1)–(A2), the hypotheses of the lemma are satisfied with $[\alpha, \beta] = [m, M]$. Since

$$L_T(\theta) = h(\text{Toe}_T(S_\theta)),$$

Eq. (3.5) gives

$$\dot{L}_T \sim \text{Cir}_T(h'(S)\dot{S}) = \text{Cir}_T(\partial_\theta \sqrt{H}). \quad (5.7)$$

Since $\text{Toe}_T(R) \sim \text{Cir}_T(R)$ and $\dot{L}_T$ is real, Lemma 2 gives

$$\dot{L}_T \text{Toe}_T(R) \dot{L}_T^\top \sim \text{Cir}_T(q), \quad q = R(\partial_\theta \sqrt{H})^2 = \frac{1}{4} S(\partial_\theta \log H)^2. \quad (5.8)$$

Let P_T^+ be the diagonal projection in Fourier coordinates onto J_T . Taking expectations in (5.6),

$$\mathbb{E} I_T^{N|Z}(Z^{(T)}) \leq 4 \text{tr}\{P_T^+ F_T \dot{L}_T \text{Toe}_T(R) \dot{L}_T^\top F_T^\dagger\}.$$

By (5.8), unitary invariance and (3.2) with $P = P_T^+$, after division by T the right-hand side differs by $o(1)$ from

$$\frac{4}{T} \text{tr}\{P_T^+ F_T \text{Cir}_T(q) F_T^\dagger\} = \frac{4}{T} \sum_{j \in J_T} q(\omega_j) \longrightarrow \frac{2}{\pi} \int_0^\pi q(\omega) d\omega,$$

where the limit is (3.3). Therefore

$$\limsup_{T \rightarrow \infty} \frac{\mathbb{E} I_T^{N|Z}(Z^{(T)})}{T} \leq \frac{1}{2\pi} \int_0^\pi S(\omega) [\partial_\theta \log H(\omega)]^2 d\omega. \quad (5.9)$$

Combining (5.4), (5.5), and (5.9),

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \frac{1}{2\pi} \int_0^\pi \{(\partial_\theta \log R)^2 + S(\partial_\theta \log H)^2\} d\omega. \quad (5.10)$$

For the choice (5.1),

$$\partial_\theta \log R = \frac{\dot{S}}{1+S}, \quad \partial_\theta \log H = \frac{\dot{S}}{S(1+S)},$$

hence pointwise

$$(\partial_\theta \log R)^2 + S(\partial_\theta \log H)^2 = \frac{\dot{S}^2}{(1+S)^2} \left(1 + \frac{1}{S}\right) = \frac{\dot{S}^2}{S(1+S)}, \quad (5.11)$$

and (5.10) becomes

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \mathcal{I}. \quad (5.12)$$

□

Proof of Theorem 1. Combine Propositions 1 and 2. □

VI. WHY (4.2) AND (5.1) ARE CHOSEN IN THE PROOFS

A. Choice of w in (4.2)

A more general lower bound on the Fisher information $I_T(\theta)$ due to Stein, Mezghani, and Nossek [3] is given by

$$I_T(\theta) \geq \dot{\mu}^\top \Sigma^{-1} \dot{\mu}, \quad \mu = \mathbb{E}_\theta(N_T), \quad \Sigma = \text{Var}_\theta(N_T),$$

but the $T \rightarrow \infty$ limit of this lower bound is nontrivial. To sidestep the issue, note that

$$\dot{\mu}^\top \Sigma^{-1} \dot{\mu} = \sup_{w \in \mathbb{R}^{|J_T|}} \frac{(w^\top \dot{\mu})^2}{w^\top \Sigma w},$$

where the ratio on the right-hand side in terms of w is precisely the right-hand side of (4.4). Now the $T \rightarrow \infty$ limit of the ratio can be taken, as per the proof of Proposition 1, and Eq. (4.2) corresponds to an asymptotically optimal choice of w to maximize the ratio.

B. Choice of R and H in (5.1)

(5.1) is also designed to optimize a more general upper bound.

Definition 1 (Admissible factorization). A family $(R_\theta)_{\theta \in \Theta}$ defines an *admissible factorization* $S_\theta = R_\theta H_\theta$ if it satisfies the hypotheses of Lemma A.1: each R_θ is real and even, $\theta \mapsto R_\theta$ is C^1 from Θ into $A(\mathbb{T})$, and $r_- \leq R_\theta(\omega) \leq r_+$ for constants $0 < r_- \leq r_+ < \infty$. We then put $H_\theta = S_\theta / R_\theta$ and define $Z^{(T)}$ and $L_T(\theta)$ by (5.2).

For an admissible factorization, $L_T(\theta) \text{Toe}_T(R_\theta) L_T(\theta)^\top = \text{Toe}_T(S_\theta)$ still holds, and Lemma A.1 gives (5.4). The matrices $\text{Toe}_T(S_\theta)$ and $\text{Toe}_T(R_\theta)$ need not commute, so (5.3) is not available. Instead, differentiate (5.2) and apply Lemma 3 twice: first with

$$a_\theta = S_\theta, \quad f(x) = x^{1/2},$$

and then with

$$a_\theta = R_\theta, \quad f(x) = x^{-1/2}.$$

The hypotheses hold with the positive intervals $[\alpha, \beta] = [m, M]$ and $[\alpha, \beta] = [r_-, r_+]$, respectively. Together with Lemma 2, the ordinary and differentiated conclusions of Lemma 3 give

$$\dot{L}_T \sim \text{Cir}_T((\partial_\theta \sqrt{S}) R^{-1/2} + \sqrt{S} \partial_\theta R^{-1/2}) = \text{Cir}_T(\partial_\theta \sqrt{H}).$$

The rest of the proof of Proposition 2 applies verbatim, with $1 + m$ replaced by r_- , up to (5.10). Since $\partial_\theta \log R = \partial_\theta \log S - \partial_\theta \log H$, this reads

$$\limsup_{T \rightarrow \infty} \frac{I_T(\theta_0)}{T} \leq \frac{1}{2\pi} \int_0^\pi \{(\partial_\theta \log S - \partial_\theta \log H)^2 + S(\partial_\theta \log H)^2\} d\omega.$$

Fix ω and write $a = \partial_\theta \log S_\theta(\omega)|_{\theta_0}$ and $u = \partial_\theta \log H_\theta(\omega)|_{\theta_0}$. The integrand

$$(a - u)^2 + Su^2$$

is strictly convex in u , with minimizer $u_* = a/(1+S)$ and minimum value $a^2 S/(1+S) = \dot{S}^2/[S(1+S)]$. Every admissible factorization with $\partial_\theta \log H_\theta|_{\theta_0} = \dot{S}/[S(1+S)]$ attains the minimum. (5.1) is an example, leading to the best upper bound at θ_0 among all admissible factorizations.

Appendix A: Finite-dimensional likelihood regularity

Fix $T \geq 3$ throughout this appendix. We verify the score identities behind (5.4) directly from (A1)–(A2).

Write $\Gamma_\theta = \text{Toe}_T(S_\theta)$. By (3.1) and (A2), $mI \leq \Gamma_\theta \leq MI$, and since Toe_T is linear with $\|\text{Toe}_T(b)\|_{\text{op}} \leq \|b\|_A$, (A1) implies that $\theta \mapsto \Gamma_\theta$ is C^1 in operator norm. Hence the $N(0, \Gamma_\theta)$ density ϕ_θ is C^1 in θ , with score

$$\text{score}_{X, \theta}(x) = \frac{1}{2} \{x^\top \Gamma_\theta^{-1} \dot{\Gamma}_\theta \Gamma_\theta^{-1} x - \text{tr}(\Gamma_\theta^{-1} \dot{\Gamma}_\theta)\}, \quad \dot{\Gamma}_\theta = \partial_\theta \Gamma_\theta, \quad (\text{A.1})$$

and on every compact subinterval $\Theta_1 \subset \Theta$,

$$\phi_\theta(x) + |\partial_\theta \phi_\theta(x)| \leq C(1 + \|x\|^2) e^{-\|x\|^2/(2M)}, \quad \theta \in \Theta_1, \quad (\text{A.2})$$

for a constant C depending on Θ_1 and T . Consequently, $\theta \mapsto \int g \phi_\theta dx$ may be differentiated under the integral sign whenever $|g|$ is bounded by a polynomial.

Let $\lambda_j^X(x) = |(F_T x)_j|^2$, the Poisson mean in (2.1) as a function of $X^{(T)} = x$. For $j \in J_T$ the zero set of λ_j^X is $\{x : \sum_t x_t \cos(\omega_j t) = \sum_t x_t \sin(\omega_j t) = 0\}$. Since $0 < \omega_j < \pi$, the vectors $(\cos \omega_j t)_{t=1}^T$ and $(\sin \omega_j t)_{t=1}^T$ are linearly independent, so this zero set is a linear subspace of codimension two and hence Lebesgue-null. The same holds for $z \mapsto \lambda_j^X(Lz)$ with any invertible L .

Lemma A.1 (Transformed experiment). *Let $(R_\theta)_{\theta \in \Theta}$ be real, even functions such that $\theta \mapsto R_\theta$ is C^1 from Θ into $A(\mathbb{T})$ and $r_- \leq R_\theta(\omega) \leq r_+$ for constants $0 < r_- \leq r_+ < \infty$; for example (5.1), with $r_- = 1 + m$ and $r_+ = 1 + M$. Let $Z = Z^{(T)} \sim N(0, \text{Toe}_T(R_\theta))$ and, conditional on $Z = z$, let the $N_{T,j}$, $j \in J_T$, be independent with*

$$N_{T,j} \mid Z = z \sim \text{Pois}(\lambda_j(\theta, z)), \quad \lambda_j(\theta, z) = |(F_T L_T(\theta) z)_j|^2 = \lambda_j^X(L_T(\theta) z),$$

where $L_T(\theta)$ is given by (5.2). Then for every $\theta \in \Theta$ the law of N_T is the original one, and

$$I_T(\theta_0) \leq I_T^Z + \mathbb{E}_{\theta_0} I_T^{N|Z}(Z).$$

Here

$$I_T^Z = \frac{1}{2} \text{tr}\{\text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R}) \text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R})\}, \quad I_T^{N|Z}(z) = \sum_{j \in J_T} \frac{\dot{\lambda}_j(z)^2}{\lambda_j(z)}$$

are the Fisher information of Z and the conditional Fisher information of N_T given $Z = z$, both at θ_0 , with $\lambda_j(z) = \lambda_j(\theta_0, z)$ and $\dot{\lambda}_j(z) = \partial_\theta \lambda_j(\theta, z)|_{\theta_0}$.

Proof. Law of N_T . Both square roots in (5.2) are real symmetric, so

$$L_T(\theta) \text{Toe}_T(R_\theta) L_T(\theta)^\top = \text{Toe}_T(S_\theta).$$

Hence $L_T(\theta)Z \sim N(0, \text{Toe}_T(S_\theta))$, and the $\lambda_j(\theta, Z)$ are exactly the intensities in (2.1) computed from $X^{(T)} = L_T(\theta)Z$.

Regularity and conditional information. By (3.1), $r_- I \leq \text{Toe}_T(R_\theta) \leq r_+ I$. Since $x^{1/2}$ and $x^{-1/2}$ are analytic on neighbourhoods of the relevant positive spectral intervals, standard analytic matrix functional calculus [8, Chapter 3] implies that $\theta \mapsto L_T(\theta)$ is C^1 in operator norm. Thus $\lambda_j(\theta, z)$ is a quadratic form in z with C^1 coefficients and, on a compact neighbourhood Θ_1 of θ_0 ,

$$\lambda_j(\theta, z) + |\partial_\theta \lambda_j(\theta, z)| \leq C\|z\|^2.$$

The Gaussian density of Z and its derivative satisfy the analogue of (A.2), with M replaced by r_+ . Together with the preceding bound, this supplies an integrable Gaussian–polynomial dominating function for the score calculations below. In particular, the usual Fisher-information chain rule and data-processing identities are applicable [4].

For Lebesgue-a.e. z , all $\lambda_j(\theta_0, z)$ are positive. The conditional Poisson score is therefore

$$\text{score}_{N|Z}(z, n) = \sum_{j \in J_T} (n_j - \lambda_j(z)) \frac{\dot{\lambda}_j(z)}{\lambda_j(z)},$$

and, by conditional independence,

$$I_T^{N|Z}(z) = \mathbb{E}_{\theta_0}[\text{score}_{N|Z}(z, N_T)^2 \mid Z = z] = \sum_{j \in J_T} \frac{\dot{\lambda}_j(z)^2}{\lambda_j(z)}.$$

With $\check{X} = F_T L_T(\theta_0)Z$ and $V = F_T \dot{L}_T Z$,

$$\dot{\lambda}_j = 2 \text{Re}(\check{X}_j^* V_j), \quad \frac{\dot{\lambda}_j^2}{\lambda_j} \leq 4|V_j|^2,$$

so $\mathbb{E}_{\theta_0} I_T^{N|Z}(Z) < \infty$. The Gaussian Fisher information is the standard covariance formula

$$I_T^Z = \frac{1}{2} \text{tr}\{\text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R}) \text{Toe}_T(R)^{-1} \text{Toe}_T(\dot{R})\}.$$

Applying the Fisher-information chain rule to the joint experiment (Z, N_T) gives [4, Lemma 1]

$$I_T^{Z,N} = I_T^Z + \mathbb{E}_{\theta_0} I_T^{N|Z}(Z).$$

Finally, N_T is the deterministic projection of (Z, N_T) onto its second component, so the Fisher-information data-processing inequality gives [4, Lemma 3]

$$I_T(\theta_0) = I_T^N \leq I_T^{Z,N}.$$

Combining the last two displays proves the asserted bound. $\square$

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